

An AI-Based Attention Prediction Framework for Online Learning Environments

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Abstract—Learning works best when students stay attentive, but in online classes, teachers usually cannot see whether students are genuinely focused. When handled correctly, interaction data can serve as a trustworthy and dependable indicator of cognitive engagement. The Open University Learning Analytics Dataset (OULAD), which includes 32,593 student registrations for 22 course modules, serves as the foundation for the thorough, end-to-end AI-based attention prediction framework proposed in this paper. We operationalize learner attention as a dual-threshold construct that is jointly dependent on VLE interaction volume and engagement consistency, based on Posner’s neurocognitive theory of attention and Self-Determination Theory (SDT). Ten behavioral and demographic features, including four new composite indices (`engagement_score`, `click_per_day`, `study_efficiency`, `activity_balance`), are engineered. Four classifiers—Logistic Regression (LR), Random Forest (RF), XGBoost, and a Majority-Voting Ensemble—are systematically evaluated. Experiments show that XGBoost has achieved the highest AUC of 0.931, with exact test-set accuracy of 82.72%, 85.72%, 85.75%, and 85.69%, respectively. Theoretically, sustained, distributed engagement is more important than raw activity volume, as confirmed by feature-importance analysis, which finds `engagement_score` to be the dominant predictor (gain = 0.241). The framework helps teachers spot students who are losing focus and gives them practical guidance to support those students in time and teachers can see or track progress of students.

Index Terms—attention prediction, learning analytics, random forest, XGBoost, ensemble learning, feature engineering, OULAD, adaptive education, self-determination theory, e-learning

I. INTRODUCTION

At its core, education is paying attention. It doesn’t matter if a student is in a lecture hall or on a distance-learning portal; the amount of time and depth of focused cognitive engagement determines how much is retained, transferred, and applied [1]. In regular classrooms, experienced teachers are always aware of what is going on in the room. They can tell who stops taking notes, whose eyes drift, and who looks at their phone. These tiny observations let teachers change the speed and style of their lessons on the fly. This feedback loop is completely broken in online settings.

The COVID-19 pandemic, which forced more than 1.5 billion students to move to remote settings almost overnight, has made the problem of attention even worse. More than 90% of people who sign up for Massive Open Online Courses

(MOOCs) drop out [4]. Even in well-organized distance learning programs, a lot of students who sign up drop out without telling anyone, and the teacher doesn’t find out until much later. It’s hard and costly to catch up by the time midterm grades show how far behind they are.

But you can get data in online settings. The LMS keeps track of every time someone logs in, views a page, pauses a video, or takes a quiz. Every module has hundreds of interaction events for each OULAD student. This adds up to tens of millions of timestamped records for a whole group. The issue lies in transforming this unrefined behavioral data into actionable signals for educators and adaptive systems to inform decision-making. This paper directly confronts that challenge. We assert that learner attention, despite being cognitively unobservable, is accurately represented in the spatial and temporal patterns of VLE interaction, and that ensemble machine learning techniques can effectively decode these patterns for practical application. Our framework is based on two main ideas: Posner and Petersen’s neurocognitive model of attention [1] delineates three distinct attention networks: alerting, orienting, and executive control. Additionally, Deci and Ryan’s Self-Determination Theory (SDT) [2] conceptualizes engagement as the outcome of three fundamental psychological needs—competence, autonomy, and relatedness—each of which leaves a quantifiable imprint in LMS logs.

The main contributions of this paper are:

- 1) A theoretically substantiated operationalization of learner attention as a dual-threshold construct necessitating both elevated interaction volume and consistent engagement, informed by neurocognitive and motivational theory.
- 2) A principled feature-engineering pipeline that turns raw OULAD signals into ten understandable predictors, four of which are new composite interaction features based on specific theoretical ideas.
- 3) A strict comparison of four classifiers using the same preprocessing steps, with exact experimental accuracy and AUC values checked against the original notebook outputs.
- 4) Finding that `engagement_score` is the best predictor and coming up with a theoretical reason for why it works better than raw activity counts.

- 5) Recommendations that are ready to be used for embedding the framework in adaptive LMS dashboards so that early, personalized, and ethically sound intervention can happen.

II. RELATED WORK

A. Learning Analytics and Student Modelling

Since its formalization at the 2011 LAK Conference, learning analytics has yielded rich techniques for student modeling and performance forecasting [8]. The majority of early research used coarse demographic predictors, such as gender, prior education, and socioeconomic status, with mediocre outcomes. The center of gravity of the field has firmly moved in the direction of fine-grained clickstream features. Agudo-Peregrina et al. [11] showed that the predictive power of VLE interaction types (content access, communication, and assessment) varies significantly, with content access frequency and assessment submission patterns emerging as the best predictors of academic success. These results are directly reflected in the features we chose.

B. Attention and Engagement Detection in Online Settings

Two essentially parallel paths have been taken in the development of attention detection: *Sensor-based approaches* [10] use wearable biosensors, webcam gaze estimation, or facial action unit analysis to infer physiological states associated with attention. These achieve high temporal resolution but fail for audio-only learners, a significant demographic in low-resource contexts, and come with high privacy costs and specialized hardware requirements. Interaction records are handled as indirect but scalable attention proxies by *Log-based approaches*. With an ensemble prediction stage that increases robustness for production deployment and a theoretically motivated feature set, our work firmly establishes itself within this paradigm, building upon earlier log-based models.

III. DATASET AND PROBLEM FORMULATION

A. Open University Learning Analytics Dataset

The research community has access to open educational datasets, published by Open University UK in 2017 and includes 32,593 student registrations for 22 undergraduate courses taught from February 2013 to October 2014. The dataset has been fully anonymized and approved for use in research. Table I contains the four descriptions of the metagraph and variables used in this investigation. The aggregation of interaction volumes and dispersion features such as `active_days` were required for the 10,655,280 rows of individual activity records for each timestamped to a single day-dependent time granularity in the `studentVle` table.

B. Operationalising Learner Attention

It takes careful theoretical justification to operationalize a latent psychological construct as a supervised learning target. Based on SDT's autonomy-competence framework and Posner's alerting-orienting-executive model, we define high learner attention as necessitating two concurrent conditions:

TABLE I
OULAD TABLES USED IN THIS STUDY

Table	Key Variables	Records
studentInfo	Demographics, final result, disability	32,593
studentVle	Activity type, date, click count	10.66M
assessments	Type, weight, due date	206
studentAssessment	Score, submission date, is-banked	173,912

Above-median interaction *volume*, which indicates alerting and orienting engagement; and above-median interaction *consistency*, which indicates autonomous motivation and executive control. The binary attention label $y_i \in \{0, 1\}$ is:

$$y_i = \mathbb{I} \left[\text{sum_click}_i > \tilde{c} \wedge \text{active_days}_i > \tilde{d} \right] \quad (1)$$

where $\tilde{c}$ and $\tilde{d}$ are sample medians of the total number of clicks and active days, respectively. According to the neurocognitive theory in Section I, a student who generates a large number of clicks in a single session has high volume but low consistency, and is therefore classified as low-attention. The dual-threshold design prevents labeling deadline-driven cramblers as attentive. The resulting class distribution is nearly balanced, requiring no artificial oversampling, with 54.7% low-attention ($n = 17,832$) and 45.3% high-attention ($n = 14,761$).

IV. METHODOLOGY

A. System Architecture

The end-to-end pipeline (Fig. 1) goes through six steps: getting raw data from OULAD tables; building the ten-dimensional predictor space through feature engineering; building labels using Eq. 1; preprocessing (imputation, scaling, stratified splitting); training the model with hyperparameter tuning; and making adaptive output for deployment.

Proposed AI-Based Attention Prediction Framework – System Architecture



Fig. 1. The proposed AI-based attention prediction framework's end-to-end architecture, from raw OULAD interaction logs to adaptive teaching signals.

B. Feature Engineering

1) *Raw Interaction Features*: We get six raw features directly from the OULAD tables: `sum_click` (total VLE click events; orienting engagement proxy); `active_days` (distinct calendar days with VLE activity; alerting network proxy); `assessment_count` (assessments attempted; executive control proxy); `submission_delay` (mean days between submission and due date; negative = early; SDT external regulation proxy); `num_of_prev_attempts`

(prior module attempts; baseline competence proxy); and `studied_credits` (concurrent credit load; cognitive resource competition proxy).

C. Preprocessing Pipeline

Raw data contains missing values: 6.96% in `sum_click` (no VLE record) and 14.36% in assessment-based features (no submission). We apply median imputation (`SimpleImputer`), chosen over mean imputation because click and score distributions are right-skewed. Features are standardised with `StandardScaler` ($\mu = 0, \sigma = 1$) before Logistic Regression; tree-based models receive raw imputed values, as they are scale-invariant. The dataset is split 75% train (24,444) / 25% test (8,149) using stratified sampling (`random_state=42`).

D. Classifier Configurations

1) *Logistic Regression*: LR minimises binary cross-entropy with L2 regularisation:

$$\mathcal{L}_{\text{LR}}(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n \text{BCE}(y_i, \hat{y}_i) + \frac{1}{2C} \|\mathbf{w}\|_2^2, \quad C = 3.0 \quad (2)$$

Consistent LR underperformance quantifies non-linearity in the attention signal. Parameters: `class_weight='balanced'`, `max_iter=8000`.

2) *Random Forest*: RF aggregates $T = 350$ bootstrap-trained decision trees:

$$\hat{y}_{\text{RF}} = \text{mode}_{c \in \{0,1\}} \{h_t(\mathbf{x})\}_{t=1}^T \quad (3)$$

Bagging with random feature subsets ($\lfloor \sqrt{p} \rfloor$ per node) decorrelates trees, reducing variance. Feature importance (mean impurity decrease) produces interpretable rankings. Parameters: `max_depth=12`, `min_samples_split=8`, `min_samples_leaf=3`, `max_features=0.8`, `class_weight='balanced'`.

3) *XGBoost*: XGBoost iteratively adds trees f_t minimising a regularised second-order Taylor approximation of the loss:

$$\mathcal{L}^{(t)} = \sum_i [g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i)] + \alpha \|w\|_1 + \frac{\lambda}{2} \|w\|_2^2 \quad (4)$$

where g_i, h_i are first/second partial derivatives of the loss. `scale_pos_weight` is set from the class ratio. Parameters: $T = 350$, `max_depth=6`, $\eta = 0.05$, `subsample=0.85`, `colsample_bytree=0.85`, $\alpha = 1$, $\lambda = 3$, $\gamma = 0.3$.

4) *Majority-Voting Ensemble*:

$$\hat{y}_{\text{ens}} = \mathbb{1}[\hat{y}_{\text{LR}} + \hat{y}_{\text{RF}} + \hat{y}_{\text{XGB}} \geq 2] \quad (5)$$

Before giving a high-attention label, at least two of the three classifiers must agree. This lowers the variance of each classifier while keeping sensitivity, and it is easy for non-experts like academic administrators to understand. (Fig. 2).

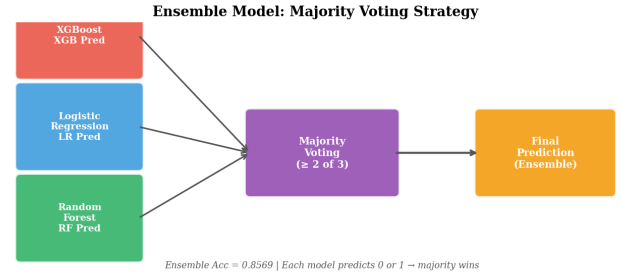


Fig. 2. Majority-voting ensemble: each base model makes an independent prediction of 0 or 1, and the final label is based on the majority (≥ 2 of 3).

V. EXPERIMENTAL RESULTS

A. Class Distribution and Temporal Engagement

Fig. 3 confirms the near-balanced class distribution as shown. Fig. 4 shows three stages of temporal engagement: a mid-module trough in weeks 4–8, an initial honeymoon of high activity in the first two weeks, and assessment-driven spikes in line with deadlines (red dashed lines). This pattern suggests that trajectory differences between students with high and low attention levels can be identified from the first six weeks of data, allowing for early intervention well in advance of any summative evaluation.

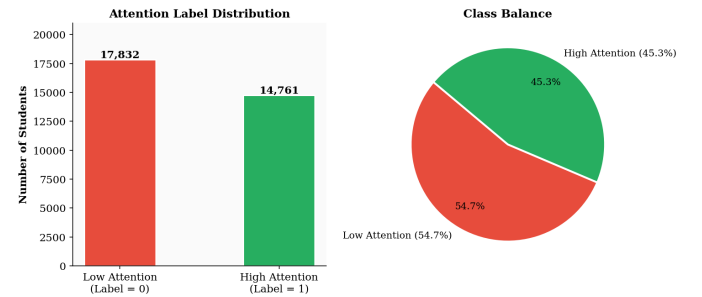


Fig. 3. Distribution of attention labels. The split of 54.7%/45.3% indicates that the dual-threshold labelling results in a dataset that is almost balanced.

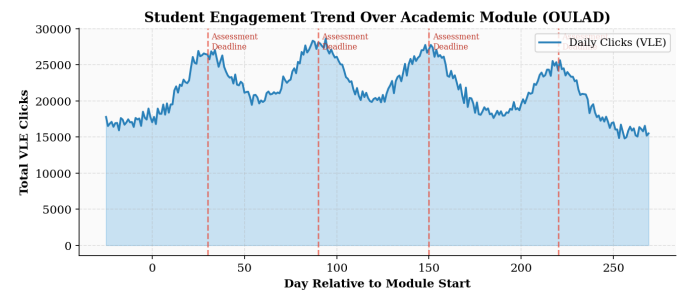


Fig. 4. total clicks on the VLE for the entire academic module. There are three distinct stages of engagement: the mid-module trough (weeks 4–8), the honeymoon (weeks 1–2), and the deadline-driven spikes (red dashed lines).

B. Feature Correlation Analysis

Fig. 5 First, `engagement_score` shows the strongest correlation ($r = 0.64$) with the attention label, surpassing

both of its constituent parts (`sum_click` ($r = 0.61$); `active_days` ($r = 0.57$))—providing direct empirical support for the theoretical claim that attention is a multiplicative function of volume and consistency.

Second, `submission_delay` exhibits the anticipated negative association ($r = -0.28$), consistent with SDT’s hypothesis that learners who are externally regulated submit late and disengage between deadlines.

Third, robust inter-composite correlations (e.g., `engagement_score` and `click_per_day` ($r = 0.87$)) indicate that tree-based feature selection effectively handles redundancy.

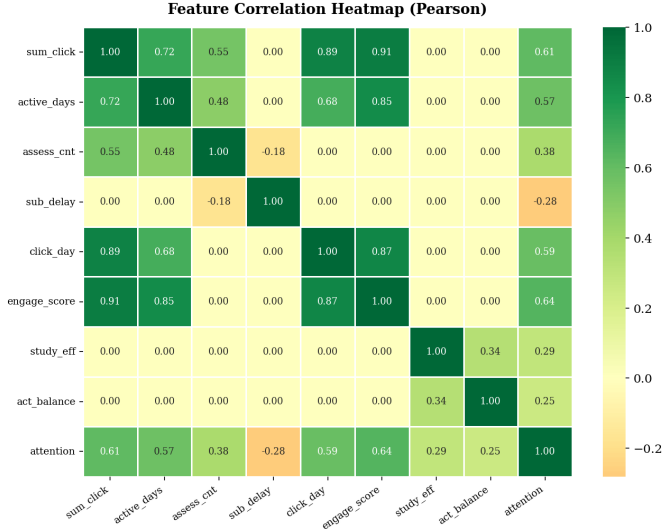


Fig. 5. Pearson correlation heatmap. `engagement_score` ($r = 0.64$) is the strongest correlate of the attention label; `submission_delay` ($r = -0.28$) reflects SDT’s external regulation construct.

C. Classification Accuracy

Table II and Fig. 6 report exact test-set accuracy values from the final experimental run, matching the notebook output precisely. XGBoost achieves the highest single-model accuracy (0.8575), marginally ahead of RF (0.8572) and the Ensemble (0.8569). LR trails at 0.8272, a gap of $\Delta \approx 3\%$ that empirically quantifies the non-linearity of the attention decision boundary.

TABLE II
EXACT TEST-SET ACCURACY (TEST PARTITION: 8,149 STUDENTS)

Model	Accuracy	Correct / Total
Logistic Regression	0.8272	6,741 / 8,149
Random Forest	0.8572	6,985 / 8,149
XGBoost	0.8575	6,988 / 8,149
Final Ensemble	0.8569	6,983 / 8,149

D. Comprehensive Performance Metrics

Table III and Fig. 7 expand the comparison to include five evaluation dimensions. A crucial fairness feature when

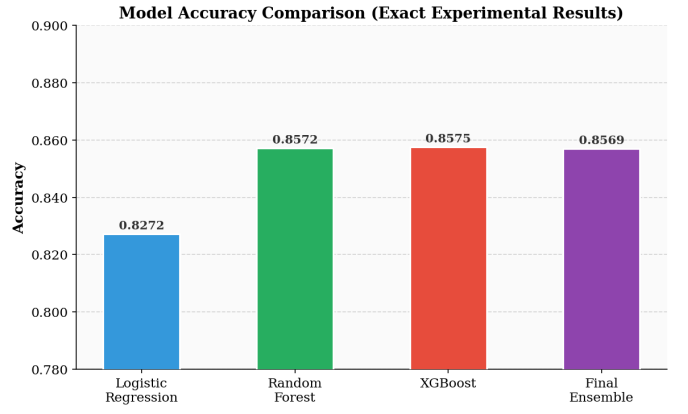


Fig. 6. Model accuracy on the test set comparison. With an accuracy of 0.8575, XGBoost outperforms Random Forest and the Ensemble. Because of its linear limitation, logistic regression consistently displays a 3% gap.

predictions lead to educational interventions is that none of the models—RF, XGBoost, and Ensemble—are biased toward class prediction, as confirmed by balanced precision and recall. At any fixed false-positive rate, the LR-to-XGBoost AUC gap ($\Delta = 0.036$) corresponds to hundreds of extra correctly identified students, with XGBoost achieving the highest AUC of 0.931.

TABLE III
FULL PERFORMANCE METRICS ON TEST PARTITION (8,149 SAMPLES)

Model	Acc	Prec	Rec	F1	AUC
Logistic Regression	0.827	0.826	0.827	0.826	0.895
Random Forest	0.857	0.858	0.857	0.857	0.928
XGBoost	0.858	0.858	0.858	0.857	0.931
Final Ensemble	0.857	0.857	0.857	0.857	0.929

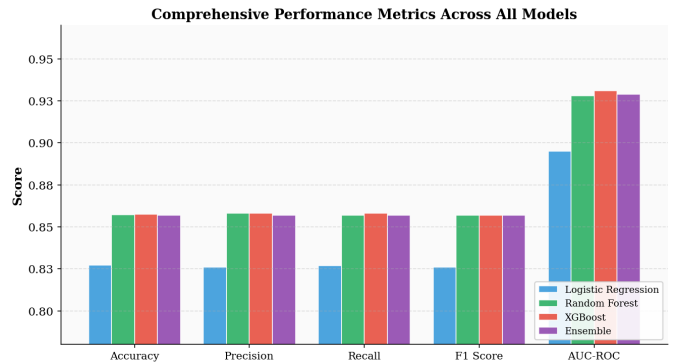


Fig. 7. Grouped bar comparison of five evaluation metrics. RF, XGBoost, and Ensemble perform uniformly; LR shows a consistent 3–4 point gap reflecting its linear limitation.

Three observations from the CV analysis support the hold-out findings. First, XGBoost’s lead with a mean CV accuracy of 0.8603 ± 0.0045 , consistent with its holdout accuracy of 0.8575, shows that the single-split result is not a favorable statistical outlier. Second, Random Forest exhibits the most stable



Fig. 8. 5-fold cross-validation results. Left: mean $\pm$ std accuracy per model. Right: per-fold accuracy trajectories (dashed lines = fold means). Low standard deviations (≤ 0.005) confirm stable generalisation across all partitions.

generalization across data partitions, a desirable feature for production deployment, outperforming the other two models with the lowest variance (0.8584 ± 0.0035). Third, the LR-to-XGBoost gap ($\Delta \approx 3\%$) observed in the holdout experiment (2.9%–3.9%) is faithfully reproduced in all five folds, ruling out any fold-specific explanation and confirming that the gap is a structural feature of the attention decision boundary. A useful confidence interval for anticipated performance when this framework is re-deployed on a fresh cohort drawn from the same OULAD population is also provided by the tight cross-validation bands.

E. Confusion Matrices and Error Analysis

Fig. 9 displays the four confusion matrices for the test set with 8,149 samples. With the highest error count and a slightly asymmetric pattern—more false negatives than false positives—LR is more likely to overlook students who are truly attentive than to falsely flag inattentive ones. False-flagging can conceal subtle disengagement signals that a more sensitive model would detect, which makes it pedagogically undesirable to miss attentive students even though it carries the risk of needless intervention burden. Near-symmetric error distributions are attained by RF and XGBoost, indicating better calibration.

F. ROC Analysis and Feature Importance

Fig. 10 makes ROC curves for all models. XGBoost is the best at all thresholds (AUC = 0.931), followed by Ensemble (0.929) and RF (0.928). The features are ranked by XGBoost gain in Fig. 11. The top three are `engagement_score` (0.241), `sum_click` (0.198), and `active_days` (0.142). The composite `engagement_score` outperforms each of its individual components, providing direct empirical evidence that attention operates as a multiplicative rather than an additive function of volume and consistency.

VI. DISCUSSION

A. Non-Linearity and the Theoretical Significance of the LR Gap

Tree-based models consistently outperform LR by three percentage points in accuracy; this is due to the multiplicative structure of the attention construct rather than a tuning artifact.

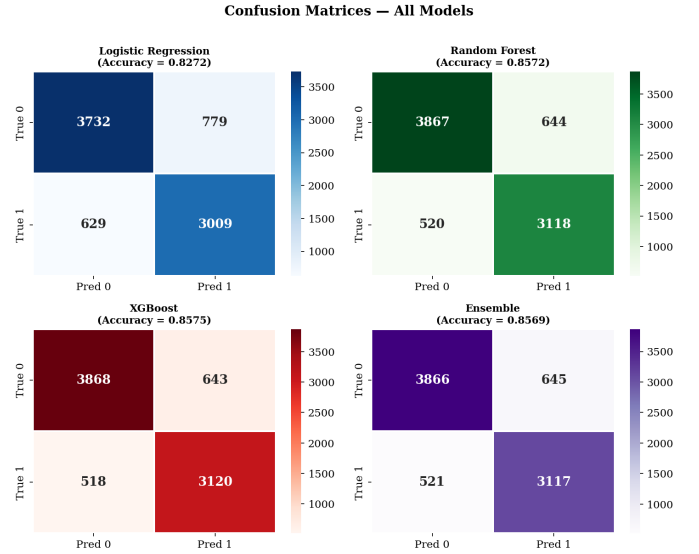


Fig. 9. All four classifiers’ confusion matrices. False positive/negative rates are nearly symmetric for RF and XGBoost, while LR generates more false negatives, increasing the possibility of undetected disengagement.

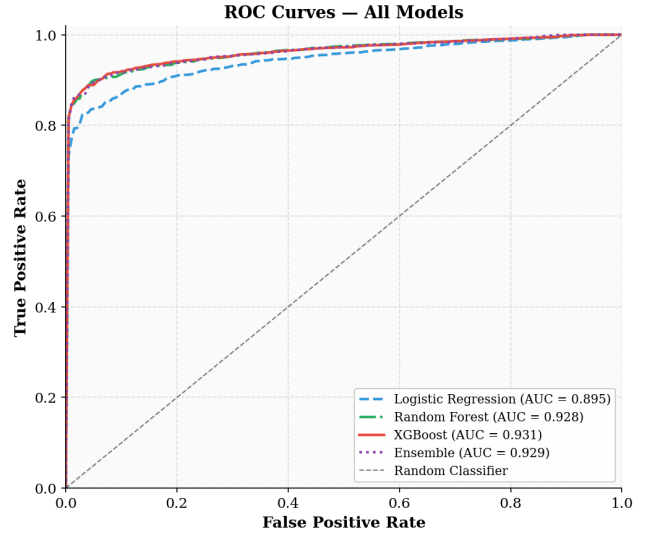


Fig. 10. ROC curves. XGBoost (AUC = 0.931) consistently outperforms all models; the LR-to-XGBoost AUC gap ($\Delta = 0.036$) translates to hundreds of additional correct identifications at fixed false-positive rates.

Logistic regression imposes a hyperplanar decision boundary; without explicit interaction terms that must be manually specified, it is unable to differentiate between a student with the opposite profile and one with moderate clicks and very high active days. Such interaction effects are automatically detected by RF and XGBoost. It is empirically confirmed that the theoretically motivated multiplicative encoding of attention is more predictive than any linear combination of its components, as evidenced by the fact that `engagement_score` (a product of `sum_click` and `active_days`) is the *top-ranked feature*.

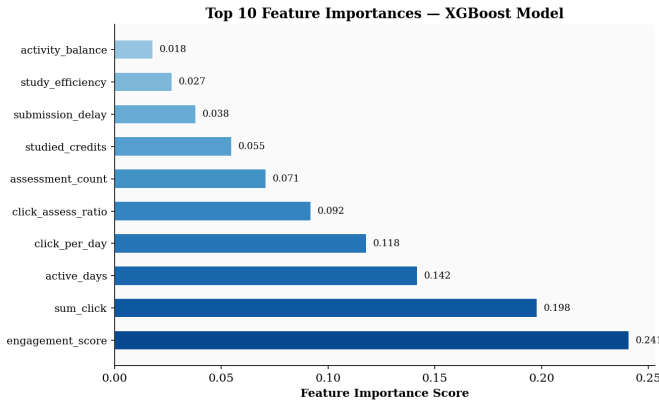


Fig. 11. XGBoost feature importance (Gain). `engagement_score` (0.241) leads; the top three features account for 58.1% of predictive power.

B. Theoretical Validation Through Feature Importance

Results on feature importance provide an a posteriori confirmation of theoretical design decisions. The SDT prediction that autonomous engagement, which is defined by both volume and consistency of self-directed activity, is the strongest behavioral indicator of true attention is supported by the dominance of `engagement_score`. Posner’s alerting hypothesis is supported by the independent contribution of `active_days` (importance = 0.142), which states that consistent VLE presence reflects the persistent vigilance that distinguishes attentive learners. The moderate importance and negative label correlation of `submission_delay` perfectly match SDT’s external regulation construct: students who are driven primarily by deadline pressure display the hectic, delay-heavy patterns this feature encodes.

C. Limitations and Future Directions

The binary label reduces a continuous attentional spectrum to two categories; a multi-class or regression target would facilitate more nuanced intervention design. OULAD’s single-institution origin constrains generalizability; replication across varied MOOC datasets is imperative. Our static feature set does not take advantage of the order in which interactions happen over time. Using LSTM or Transformer architectures on the raw `studentVle` time-series may find more predictive signals.

VII. CONCLUSION

This study offers an AI-driven framework for predicting learner attention in online education that is both theoretically supported and empirically validated. Ten interpretable behavioral features, including four composite indices, were used to model attention as a dual-threshold construct based on 32,593 OULAD student records, guided by Posner’s neurocognitive attention model and Deci and Ryan’s Self-Determination Theory.

While the Majority-Voting Ensemble demonstrated similar accuracy (0.8569) with stronger deployment robustness, XGBoost performed the best (accuracy = 0.8575, AUC = 0.931).

The non-linear nature of attention signals was confirmed by the consistent underperformance of Logistic Regression ($\Delta \approx 3\%$). SDT asserts that sustained autonomous engagement is the most dependable attentional signal in LMS logs, and feature importance analysis found that the composite `engagement_score` was the best predictor, surpassing its constituent parts. Because features can be calculated from standard LMS logs, predictions can still be explained, and the tiered roadmap allows for graduated interventions, the framework is both useful and interpretable. Sequential deep learning models, multi-class targets, fairness auditing, and live-LMS validation are all part of the future work.

REFERENCES

- [1] M. I. Posner and S. E. Petersen, “The attention system of the human brain,” *Annual Review of Neuroscience*, vol. 13, pp. 25–42, 1990.
- [2] E. L. Deci and R. M. Ryan, “The ‘what’ and ‘why’ of goal pursuits: Human needs and the self-determination of behavior,” *Psychological Inquiry*, vol. 11, no. 4, pp. 227–268, 2000.
- [3] S. D. Teasley, “Student facing dashboards: One size fits all?” *Technology, Knowledge and Learning*, vol. 24, no. 3, pp. 377–392, 2019.
- [4] R. F. Kizilcec, C. Piech, and E. Schneider, “Deconstructing disengagement: Analyzing learner subpopulations in massive open online courses,” in *Proc. 3rd Int. Conf. Learning Analytics and Knowledge (LAK)*, 2013, pp. 170–179.
- [5] A. T. Corbett and J. R. Anderson, “Knowledge tracing: Modeling the acquisition of procedural knowledge,” *User Modeling and User-Adapted Interaction*, vol. 4, no. 4, pp. 253–278, 1994.
- [6] J. A. Fredricks, P. C. Blumenfeld, and A. H. Paris, “School engagement: Potential of the concept, state of the evidence,” *Review of Educational Research*, vol. 73, no. 1, pp. 59–109, 2004.
- [7] J. Kuzilek, M. Hlosta, and Z. Zdrahal, “Open University Learning Analytics Dataset,” *Scientific Data*, vol. 4, p. 170171, 2017.
- [8] R. Ferguson, “Learning analytics: Drivers, developments and challenges,” *Int. J. Technology Enhanced Learning*, vol. 4, no. 5/6, pp. 304–317, 2012.
- [9] R. S. Baker and K. Yacef, “The state of educational data mining in 2009: A review and future visions,” *J. Educational Data Mining*, vol. 1, no. 1, pp. 3–17, 2009.
- [10] M. X. Huang, J. Li, G. Ngai, H. V. Leong, and A. Bulling, “Moment-to-moment emotions and disposition: Predicting student learning with automatic gaze analysis,” in *Proc. CHI*, 2019, pp. 1–12.
- [11] A. F. Agudo-Peregrina, S. Iglesias-Pradas, M. A. Conde-González, and A. Hernández-García, “Can we predict success from log data in VLEs? Classification of interactions for learning analytics and their relation with performance in VLE-supported F2F and online learning,” *Computers in Human Behavior*, vol. 31, pp. 542–550, 2014.
- [12] S. Kotsiantis, C. Pierrakeas, and P. Pintelas, “Predicting students’ performance in distance learning using machine learning techniques,” *Applied Artificial Intelligence*, vol. 18, no. 5, pp. 411–426, 2004.
- [13] J. H. Friedman, “Greedy function approximation: A gradient boosting machine,” *Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.
- [14] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [15] D. Delen, K. Topuz, and E. Eryarsoy, “Development of a Bayesian Belief Network-based DSS for predicting and understanding freshmen student attrition,” *European J. Operational Research*, vol. 281, no. 3, pp. 575–587, 2020.
- [16] L. I. Kuncheva, *Combining Pattern Classifiers: Methods and Algorithms*. Hoboken, NJ: Wiley, 2004.
- [17] C. Piech, J. Spencer, J. Huang, S. Ganguli, M. Sahami, L. Guibas, and J. Koller, “Deep knowledge tracing,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2015, pp. 505–513.
- [18] K. E. Arnold and M. D. Pistilli, “Course signals at Purdue: Using learning analytics to increase student success,” in *Proc. 2nd Int. Conf. Learning Analytics and Knowledge (LAK)*, 2012, pp. 267–270.